

Extraneous Solutions and Real-World Applications - Practice

Algebra II | Unit 2 | Lesson 6 Practice | TEK A2.6.D, A2.6.E

Objective: I can identify extraneous solutions when solving absolute value equations with a variable on both sides.

Section 1 — Recall

1. Is $x = 4$ a valid or extraneous solution to $|2x - 3| = x + 1$?

$|5| = 5$, $4 + 1 = 5$. $5 = 5 \rightarrow$ _____

2. Is $x = -1$ a valid or extraneous solution to $|x + 5| = 2x + 10$?

$|4| = 4$, $2(-1) + 10 = 8$. $4 \neq 8 \rightarrow$ _____

Section 2 — Apply

3. Solve $|x - 3| = 2x$, checking both solutions.

$x =$ _____ or $x =$ _____ Solution set: {_____}

4. Solve $|2x + 3| = x + 9$, checking both solutions.

$x =$ _____ or $x =$ _____ Solution set: {_____} (both valid)

Section 3 — Challenge

5. Solve $|3x - 2| = 5x + 10$, checking both solutions.

$x =$ _____ or $x =$ _____ Solution set: {_____}

6. Solve $|x + 7| = -2x - 1$, checking both solutions.

$x =$ _____ or $x =$ _____ Solution set: {_____}

Section 4 — Explain It

7. Explain why checking a solution in the ORIGINAL equation matters, even after correctly splitting and solving. Use one of today's problems as your example.
